LAB-13: Statistical Computing (MTH210)

Perform the following:

1. Complete the work of Lab-12.

2. Suppose U_1, U_2 and U_3 are three independent exponential random variables with parameters $\lambda_1 = 0.15$, $\lambda_2 = 0.16$ and $\lambda_3 = 0.18$, and $X = \min\{U_1, U_3\}$ and $Y = \min\{U_2, U_3\}$. Generate a random sample of size $N = 25$ of (X, Y). Based on the generated sample compute

   - (1) the MLEs of λ_1, λ_2 and λ_3 by using a three dimensional optimization process
   - (2) by using EM algorithm.

   Repeat the process 1000 times and compute the average estimates of the MLEs and the associate mean squared errors based on 1000 replications in both the cases. Repeat the process for $N = 50$ and $N = 75$ and also for $\lambda_1 = 0.2$, $\lambda_2 = 0.2$ and $\lambda_3 = 0.25$.